

ANSI Stroop: Probing Inhibitory Control in Small Local LLMs

Alessandra Bagnato

Softteam, Docaposte Groupe

alessandra.bagnato@docaposte.fr

The Stroop task, a classic measure of executive control in cognitive psychology, asks participants to name the ink color of a word while ignoring its semantic content. This work adapts that paradigm to large language models (LLMs), examining whether small, locally run models exhibit analogous interference effects when word meaning and color-coding signals conflict.

1. Motivation and Problem

The Stroop task [1] asks a subject to name the ink color of a word while ignoring its meaning, a classic test of inhibitory control.

Do language models show an analogous conflict between a symbolic code and a high-frequency word?

- Patel, Wang and Fan [3] showed GPT-4o and Claude 3.5 Sonnet name the written word rather than the rendered color.
- This work targets small, open-weight models running CPU-only on consumer hardware.

Approach in one line

Visual color replaced by ANSI escape codes injected as tokens, four prompt variants.

Prompt excerpt (Baseline)

```
INSTRUCTION: For each item below, identify the INK COLOR
(the ANSI color code applied), NOT the word written.
Items:
1. \x1b[31mMAGENTA\x1b[0m   [code ANSI: \x1b[31m]
2. \x1b[34mROUGE\x1b[0m      [code ANSI: \x1b[34m]
```

3. Method / System Overview

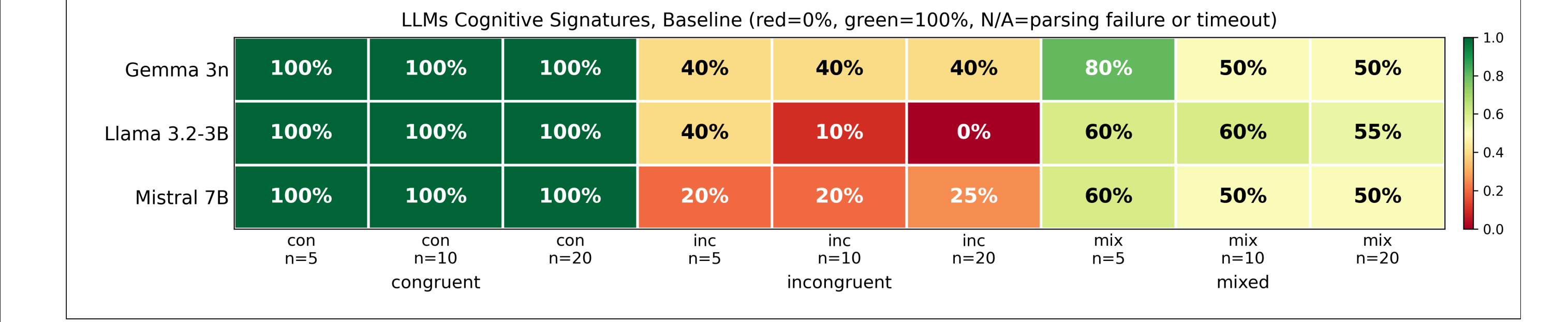
All runs are local, CPU-only, through LM Studio, temperature 0.0, fixed seed, following digital sobriety [4].

Model	Params	Quant.	Rationale
Gemma-3n-E4B	6.9B	Q4_K_M	strong instruction following
LLaMA-3.2-3B-Instruct	3B	Q4_K_M	smallest model
Mistral-7B-Instruct-v0.3	7B	Q4_K_M	open-weight baseline

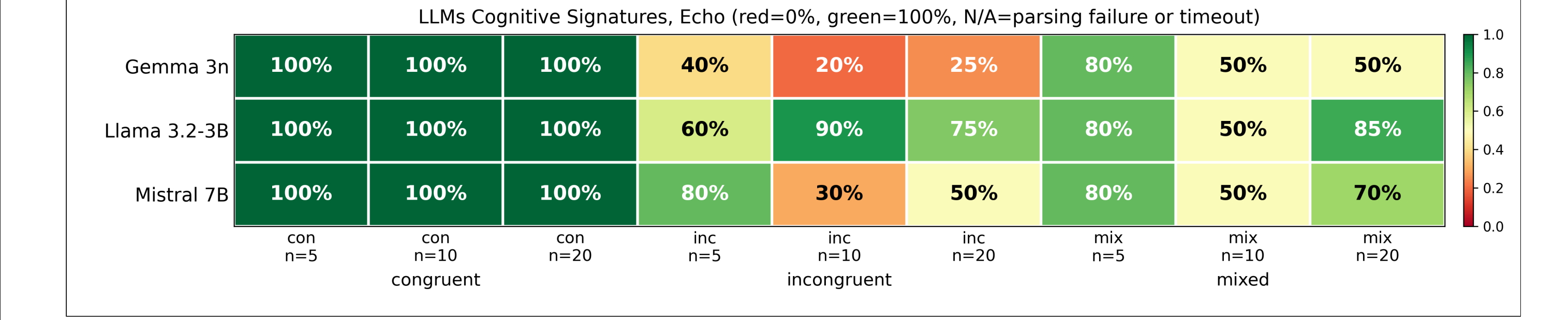
Four variants

- Baseline** (Variant A): raw ANSI code plus reference table
- Echo** (Variant B): codes wrapped in Bash echo -e syntax
- Ligurian**: Echo format, rare color names, incongruent only
- C3**: explicit Stroop warning plus required chain-of-thought

Accuracy heatmap, Baseline variant, n=5/10/20.



Accuracy heatmap, Echo variant, n=5/10/20.



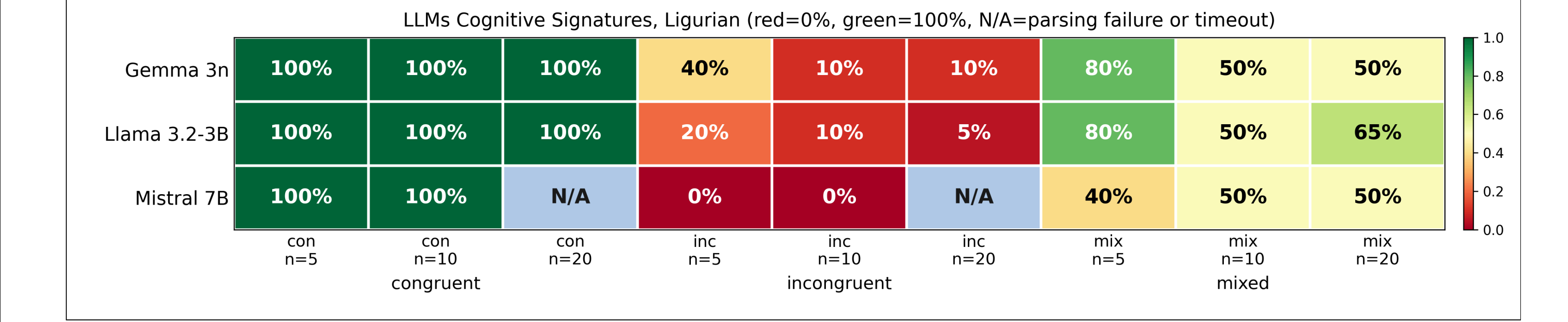
5. Benchmark and Results

Incongruent accuracy at n=20, by variant and model.

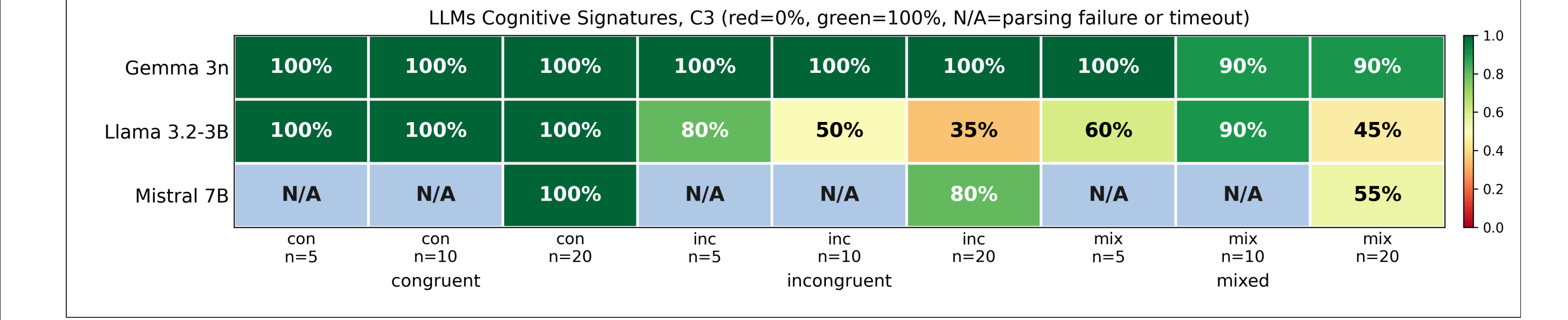
Model	Baseline	Echo	Ligurian	C3
Gemma 3n	40%	25%	10%	100%
Llama 3.2-3B	0%	75%	5%	35%
Mistral 7B	25%	50%	n/a	80%

C3 raises Gemma's incongruent accuracy from 40% to 100%, with "However" in 10 of 36 reasoning chains, correct in all 10.

Accuracy heatmap, Ligurian variant, n=5/10/20.



Accuracy heatmap, C3 variant, n=5/10/20.



Keywords: Stroop effect, inhibitory control, local LLMs, ANSI codes, prompt framing

2. Background

Kahneman's dual process theory [2] maps reading onto System 1 (fast, automatic) and deliberate override onto System 2. In a model, next-token prediction plays System 1, chain-of-thought reasoning plays System 2.

- MacLeod's review [5] attributes human Stroop interference to reading automaticity.
- Plato's Ion [12]: a rhapsode fluent on Homer, helpless on other poets, fluency without principled understanding.
- The Ligurian palette applies this test: rare compound English names on seven consecutive 256-color codes, outside any vocabulary a model would have memorized.

Standard palette

```
\033[31m = rouge (red)
\033[32m = vert (green)
\033[33m = jaune (yellow)
\033[34m = bleu (blue)
\033[35m = magenta
\033[36m = cyan
\033[38;5;76m = chartreuse
               (bright yellow-green)
```

Ligurian palette

```
\033[38;5;73m = lagoon-teal
\033[38;5;74m = sea-green
\033[38;5;75m = sky-teal
\033[38;5;76m = azure-green
\033[38;5;77m = aqua-cyan
\033[38;5;78m = emerald-cyan
\033[38;5;79m = mint-green
```

4. Technical Detail: Palettes and Prompts

Stimulus formats

Each item is a Bash echo -e command; the model must name the ink color (the code), not the printed word.

```
Variant A:  MAGENTA [ANSI code: \x1b[35m]
Variant B:  echo -e "\e[35mMAGENTA\e[0m"
Liguria:    echo -e "\e[38;5;74mSEA-GREEN\e[0m"
```

Key components

- Reference table** (Variant A): the code-to-color mapping is given, so failures are not palette ignorance.
- C3 warning**: names the interference before it occurs and says to trust the code.

Why reasoning matters

The warning alone is not the active ingredient: it comes with a required reasoning chain. Gemma's three steps (decode the code, note the word, apply the override) reach 100% accuracy only when the chain is completed. The reasoning is the mechanism of the override.

Example prompt (C3, header)

```
WARNING: This task contains a deliberate cognitive
interference effect known as the Stroop effect. The
written word will automatically pull your attention
toward its own meaning. This is TRAP. The word is
IRRELEVANT. Your ONLY source of truth is the ANSI
color code before the word. Resist the interference.
Focus exclusively on the ANSI code.
```

Output format

Each answer is parsed as JSON and scored. For the standard palette, an alias table (e.g. red → rouge) normalizes English variants, so they are not counted as invalid. Ligurian names are already English.

```
{
  "word": "ROUGE",
  "correct_answer": "bleu",
  "reasoning": "The code is \e[34m, which maps to bleu (blue). However, the word 'ROUGE' is
printed in red. The color code overrides the word's inherent color.",
  "ink_color": "bleu",
  "correct": true
}
```

6. Discussion and Conclusion

Three inhibitory mechanisms

Schema-based inhibition (Gemma, C3); token-sequence matching without grounding (Llama, Echo); word-pathway dominance tied to corpus frequency (Baseline, Ligurian).

The role of explicit reasoning

Necessary but not sufficient. Gemma's chain verbalizes System 2 and reaches 100% at n=20. Llama's chain echoes the code, then reverts to the word in about half the incongruent items: stating the schema is not following it.

Llama, C3, incongruent item: the reasoning echoes the code, the answer reverts to the word.

```
{
  "word": "ROUGE",
  "correct_answer": "bleu",
  "reasoning": "\e[34m",
  "ink_color": "rouge",
  "correct": false
}
```

Limitations

- Single random seed, ordering robustness not evaluated
- Mistral's C3 format instability may reflect template handling

Take-away

All three small local models show human-like Stroop interference, driven by corpus frequency rather than architecture alone. Explicit metacognitive framing can largely overcome it in some models. Future work: multiple seeds, frequency versus language, longer lists, other symbolic domains.



[1] Stroop, J.R.: Studies of interference in serial verbal reactions. J. Exp. Psychol. 18(6), 643-662 (1935). [2] Kahneman, D.: Thinking, Fast and Slow. Farrar, Straus and Giroux (2011). [3] Patel, S.C., Wang, H., Fan, J.: Deficient executive control in transformer attention. PNAS Nexus 5(6), pgag149 (2026). [4] P  r  a, C., G  rard, J., de Benedittis, J.: Digital sobriety. Technol. Forecast. Soc. Change 194, 122670 (2023). [5] MacLeod, C.M.: Half a century of research on the Stroop effect. Psychol. Bull. 109(2), 163-203 (1991). [12] Plato: Ion. Transl. P. Woodruff. In Plato: Complete Works, 937-949. Hackett (1997).